

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- GenAI Forward Deployed Engineer and software professional with **5+ years of experience** building production-grade applications using Python, cloud platforms, APIs, data pipelines, and AI systems, with hands-on ownership from technical discovery and rapid prototyping through customer deployment and launch.
- Experienced architecting and delivering **enterprise AI solutions on GCP**, including Vertex AI, Gemini models, agentic workflows, multi-agent systems, RAG architectures, vector databases, structured and unstructured data pipelines, and customer-specific integrations.
- Built production-grade **agentic applications** using LangGraph, CrewAI, ADK concepts, ReAct, self-reflection, hierarchical delegation, tool calling, MCP servers, state management, and orchestration patterns for complex enterprise workflows.
- Strong background in **Python software development, FastAPI, REST APIs, microservices, SQL, PySpark, cloud-native architecture, data modeling, legacy system integration, and secure enterprise connectivity** across regulated industries.
- Led customer-facing **technical discovery sessions, architecture workshops, solution design, rapid prototypes, white-glove deployments, and co-build engagements**, translating business challenges into bespoke AI systems that delivered measurable business value and adoption.
- Designed **evaluation pipelines and observability frameworks** tracking accuracy, safety, latency, granular traces, tokens/sec, cost-per-request, model behavior, state transitions, and end-user feedback to improve production reliability and performance.
- Applied a high-agency builder mindset to identify integration blockers, data-readiness issues, security constraints, and repeatable field patterns, converting successful approaches into **reusable modules, engineering best practices, and product feature feedback**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Engineering:** Python, JavaScript, TypeScript, Java, FastAPI, Flask, REST APIs, Microservices, SQL, PySpark, Object-Oriented Programming, Distributed Systems, Production-Grade Software Development.
- **Google Cloud & AI Platforms:** GCP, Vertex AI, Gemini, Google Cloud Storage, BigQuery, Cloud Run Concepts, Cloud Functions, IAM, VPC/Networking Concepts, Cloud Logging, Cloud Monitoring, AI Platform Architecture.
- **GenAI & Agentic Systems:** LLMs, Multi-Agent Systems, LangGraph, CrewAI, ADK, ReAct, Self-Reflection, Hierarchical Delegation, MCP Servers, Tool Calling, Function Calling, Agent Orchestration, State Management.
- **RAG & Data Engineering:** Retrieval-Augmented Generation, Vector Databases, Embeddings, Semantic Search, Structured Data, Unstructured Data, Data Pipelines, ETL/ELT, Data Modeling, Document Processing, Retrieval Pipelines.
- **Evaluation & Observability:** AI Evaluation Pipelines, Accuracy, Safety, Latency, Granular Tracing, Tokens/sec, Cost-per-Request, State Tracing, Agent Observability, Model Monitoring, Logging, User Feedback, Performance Optimization.
- **Enterprise Integration & Security:** APIs, Legacy Data Silos, Enterprise Systems, OAuth2/OIDC, JWT, SAML, IAM, RBAC, Security Perimeters, Authentication, Authorization, Data Privacy, Secure Cloud Integration.
- **FDE & Delivery:** Technical Discovery, Customer Co-Building, Rapid Prototyping, Production Launch, White-Glove Deployment, Architecture Design, Reusable Modules, Product Feedback Loops, Engineering Best Practices, Customer Adoption, ROI-Focused Delivery.

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Architected and delivered **production-grade GenAI and agentic solutions** using GCP, Vertex AI, Gemini, Palantir, Python, FastAPI, REST APIs, and enterprise data services, taking customer use cases from discovery and rapid prototype through deployed operational workflows.
- Led **technical discovery sessions** with business stakeholders, architects, and engineering teams to identify integration blockers, data-readiness gaps, security constraints, operational requirements, and measurable ROI before defining customer-specific AI architectures.
- Built **multi-agent systems** using LangGraph, CrewAI, ADK concepts, ReAct, self-reflection, hierarchical delegation, tool calling, MCP servers, RAG, embeddings, vector databases, and governed enterprise data to execute complex insurance and operational workflows.
- Engineered structured and unstructured data pipelines using **Python, PySpark, SQL, Palantir Foundry, vector retrieval, document processing, APIs, and enterprise data sources**, connecting AI applications with live infrastructure and legacy data silos.
- Designed high-performance **evaluation and observability pipelines** to measure accuracy, safety, latency, tokens/sec, cost-per-request, retrieval quality, state transitions, granular traces, tool behavior, and user feedback across production agentic workflows.
- Implemented secure cloud and enterprise integrations using **GCP IAM, APIs, OAuth2/OIDC, JWT, SAML, RBAC, security perimeters, Docker, Terraform, CI/CD, logging, and monitoring**, enabling reliable deployment into customer environments.
- Identified repeatable field patterns and platform friction points, packaging successful approaches into **reusable Python modules, agent components, prompt patterns, MCP integrations, architectural templates, engineering documentation, and product feature feedback** for broader engineering reuse.

Environments: GCP, Vertex AI, Gemini, Palantir Foundry/AIP, Python, FastAPI, LangGraph, CrewAI, ADK Concepts, ReAct, MCP Servers, RAG, Vector Databases, PySpark, SQL, REST APIs, OAuth2/OIDC, JWT, Docker, Terraform, CI/CD, Evaluation Pipelines, Observability, Granular Tracing.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and delivered cloud-native healthcare applications using **Python, FastAPI, REST APIs, AWS/GCP services, SQL, and microservices**, supporting secure production workloads and customer-facing enterprise workflows.
- Built **LLM-powered and agentic applications** using Gemini/OpenAI-style models, LangGraph patterns, RAG, embeddings, vector databases, tool calling, state management, and multi-step orchestration for healthcare information and operational use cases.
- Developed structured and unstructured data pipelines integrating **documents, APIs, relational data, enterprise services, and vector retrieval**, enabling grounded AI responses and reliable downstream business actions.
- Architected secure integrations across **REST APIs, identity services, legacy applications, OAuth2/OIDC, JWT, IAM, RBAC, and cloud security controls**, balancing usability, privacy, compliance, and production deployment requirements.
- Implemented **evaluation, tracing, and observability** for AI and backend services, measuring output quality, latency, token usage, cost, failures, state transitions, and service health through structured logs and operational dashboards.
- Collaborated directly with product owners, architects, and engineering stakeholders through **technical discovery, architecture design, prototype reviews, deployment planning, and production rollout**, resolving integration and data-readiness blockers.
- Converted recurring implementation patterns into **reusable Python libraries, API clients, prompt templates, retrieval components, deployment guides, architecture notes, and engineering runbooks** to accelerate subsequent delivery work.

Environments: Python, FastAPI, REST APIs, GCP/AWS, Gemini/OpenAI APIs, LangGraph Concepts, RAG, Embeddings, Vector Databases, SQL, Microservices, OAuth2/OIDC, JWT, IAM, Docker, CI/CD, Structured Logging, Tracing, Evaluation Pipelines, Observability.

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Designed and developed production enterprise systems using **Python, FastAPI, Flask, REST APIs, SQL, PySpark, and microservices**, supporting scalable business workflows and distributed applications across customer environments.
- Built reusable backend services for **workflow orchestration, data processing, API integration, asynchronous operations, business-rule execution, and state management**, enabling resilient communication across enterprise platforms.
- Engineered data pipelines for **structured and unstructured data** using Python, PySpark, SQL, ETL/ELT, PostgreSQL, MySQL, Oracle, document processing, and retrieval-oriented patterns for downstream analytical and application use cases.
- Integrated enterprise applications with **legacy data silos, REST APIs, distributed services, authentication layers, and event-driven patterns**, addressing data compatibility, service reliability, and integration complexity.
- Implemented application observability through **structured logging, performance metrics, tracing, latency analysis, exception monitoring, and operational dashboards**, improving production diagnostics and engineering response times.
- Partnered with architects, product teams, and business stakeholders during **technical discovery, architecture design, implementation planning, trade-off analysis, deployment, and production support** for customer-facing solutions.
- Created reusable engineering assets including **Python libraries, API abstractions, data-processing components, architecture templates, deployment procedures, documentation, and operational runbooks** to standardize delivery across teams.

Environments: Python, FastAPI, Flask, PySpark, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Structured/Unstructured Data, Event-Driven Architecture, Docker, Jenkins, Git, CI/CD, Logging, Tracing, Cloud-Native Applications.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed production backend applications using **Python, Flask, REST APIs, SQL, and microservices architecture**, supporting distributed enterprise systems and customer-facing operational workflows.
- Built reusable Python components for **business logic, workflow orchestration, state handling, data transformation, API communication, configuration management, and asynchronous processing** across enterprise applications.
- Engineered structured data-processing pipelines using **Python, SQL, PostgreSQL, MySQL, batch workflows, scheduled jobs, and data modeling**, supporting high-volume transactional and operational datasets.
- Integrated web and backend applications with **REST APIs, legacy enterprise services, relational databases, authentication systems, and external platforms**, addressing connectivity, security, and data transformation requirements.
- Applied secure software engineering patterns including **authentication, authorization, API contracts, Git, Docker, CI/CD, Linux, structured logging, monitoring, and environment configuration** to improve deployment reliability.
- Worked with senior engineers, architects, analysts, and customer stakeholders through **technical discovery, requirements clarification, architecture discussions, implementation planning, production deployment, and post-launch support**.
- Produced reusable **service modules, API patterns, technical documentation, architecture notes, deployment instructions, troubleshooting guides, and engineering handoff materials** to improve maintainability and knowledge transfer.

Environments: Python, Flask, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Data Modeling, Batch Processing, Legacy Integrations, Docker, Git, CI/CD, Linux, API Security, Structured Logging, Monitoring, Distributed Systems.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.